SAFE MODERATE HIGH

EXHAUSTSHIELD

Stop the AI Exhaust Leak — Before the Prompt Ever Leaves

A pilot in intelligent risk-routing, secret redaction, and human-verified knowledge protection for enterprise AI gateways

DESIGN

Manikandan

STATUS

Pilot Lab — Live

THE WARNING BEHIND THE PILOT

> *A hard boundary across which nothing crosses — not even the intelligence exhaust — without consent.*
>
> **— Satya Nadella, Chairman & CEO, Microsoft**

Every prompt, correction, and agent action an enterprise feeds into external AI becomes "exhaust" — a quiet trail of institutional know-how that leaks out, trace by trace.

The Risk

Sensitive prompts flow to external AI with no visibility or control.

The Pilot

ExhaustShield routes every prompt through a 3-lane risk gateway before it leaves.

The Proof

Live in a contained lab, tested against real leak scenarios end to end.

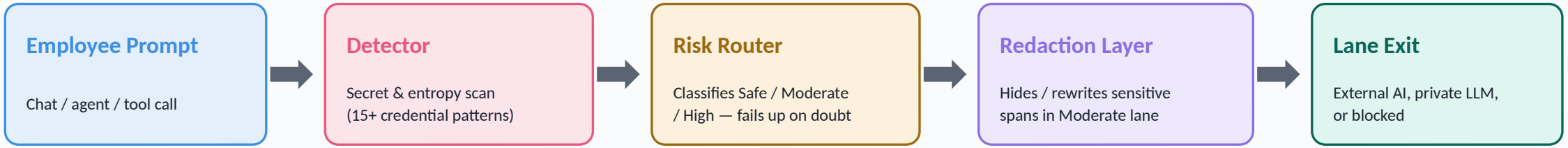
PAIN POINT → SOLUTION

Closing the gap between employee productivity and data protection

PAIN	SOLUTION
Employees paste secrets, keys & PII straight into external AI chat tools.	Detector scans every prompt for 15+ credential & PHI/PII patterns before egress.
Business-sensitive figures (deals, pricing, revenue) leak in plain prompts.	Moderate lane auto-redacts & rewrites sensitive spans, then allows external AI.
No visibility into what left the building or was blocked.	Every decision is written to an immutable local audit ledger, tagged by lane.
Wrong AI answers get repeated with no institutional memory.	Human-reviewed corrections are stored locally and reused — never sent to train external models.

ARCHITECTURE FLOW

Every prompt earns its exit through a 3-lane risk gateway



SAFE

Clean prompt → sent to external AI as-is (one-click confirm).

MODERATE

Business specifics hidden/rewritten → then external AI.

HIGH

Credentials, PHI, hard IDs → private LLM only. Never external.

PROTECTION & MEMORY LAYER (local-only)

- Immutable audit ledger — every route, redaction, and block time-stamped on disk
- Human review queue for unanswered / disputed questions, corrected once, reused forever
- Local knowledge base of approved policy answers — zero data ever sent to train external models

WHY IT'S DIFFERENT

ExhaustShield vs. the status quo of open AI access

	STATUS QUO	EXHAUSTSHIELD
Prompt Visibility	✕ No record of what employees send externally	✓ Every prompt classified and logged before it leaves
Secret Handling	✕ Keys & credentials pasted in as plain text	✓ Detected and redacted pre-egress, automatically
Sensitive Business Data	✕ Deal terms and figures shared freely	✓ Auto-rewritten in the Moderate lane before sending
Institutional Memory	✕ Corrections re-taught to external models each time	✓ Corrections stored locally, reused, never exported
Data Residency	✕ Unclear where prompts and outputs end up	✓ High-risk data never leaves the private LLM boundary

UNDER THE HOOD

Six categories of exhaust the detector is built to catch

Cloud & API Credentials

AWS, OpenAI, Anthropic, GitHub, Slack keys & tokens

Regulated People-Data

SSNs, employee/contractor IDs, PHI-shaped identifiers

Auth & Session Tokens

Bearer tokens, JWTs, OAuth secrets in transit

Business Financials

Deal values, pricing tiers, revenue figures

Obfuscated Secrets

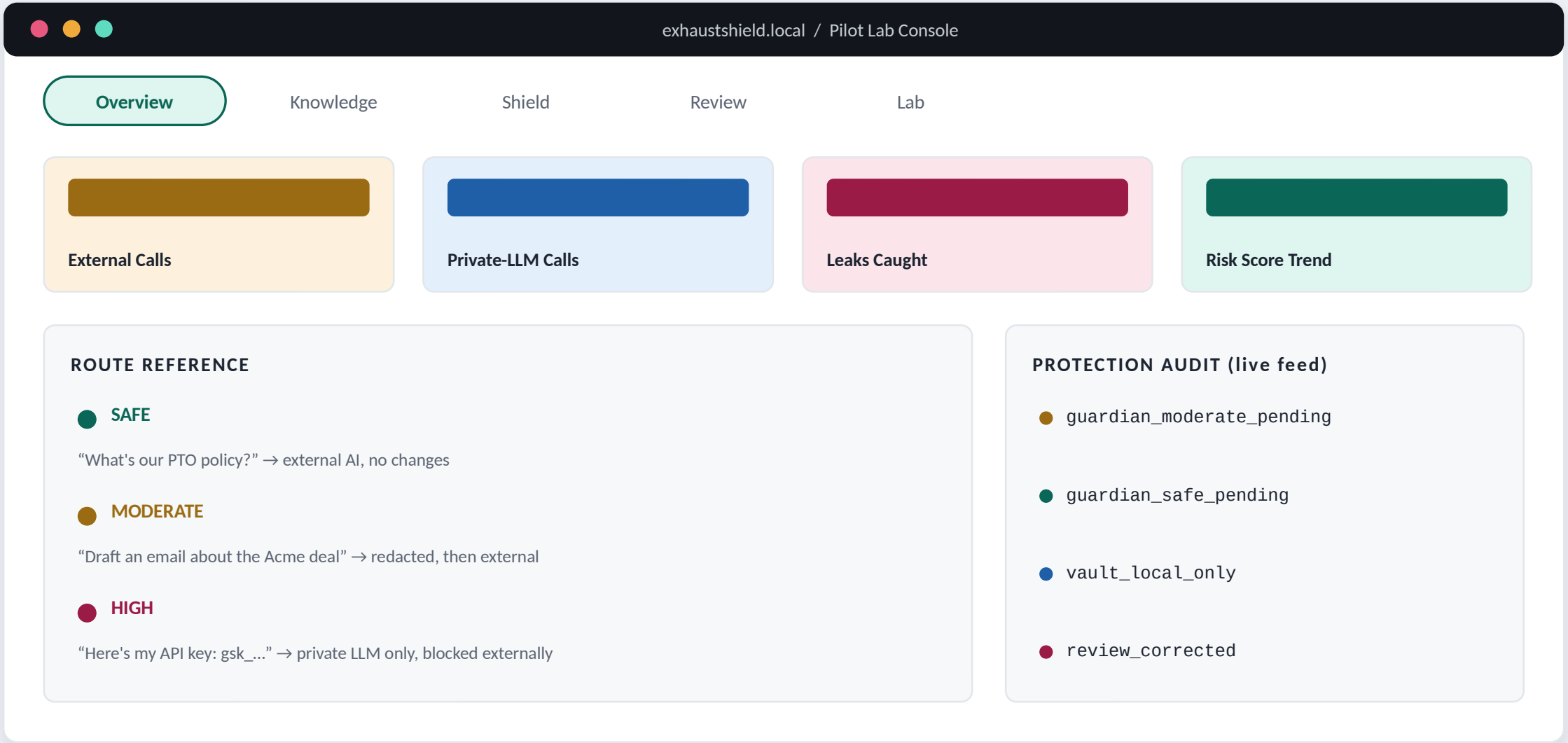
Spaced, dashed, or base64-hidden credential strings

Payment Instruments

Card numbers and other hard financial identifiers

PRODUCT WALKTHROUGH

Inside the ExhaustShield console



SCALING THE MODEL

From contained pilot lab to enterprise-wide production

CURRENT PILOT

Contained Lab Deployment

- Single gateway instance, one team's traffic
- Detector rules tuned on a handful of sample scenarios
- Local knowledge base seeded with a few sample policy docs
- Manual review queue checked ad hoc
- Audit ledger on local disk, single machine

PRODUCTION SCALE

Enterprise-Wide Rollout

- Load-balanced gateway cluster across every department's traffic
- Detector patterns continuously trained on org-wide flagged events
- Knowledge base synced to the full enterprise policy & SOP library
- Structured review workflow with SLA-based routing to owners
- Centralized, tamper-evident audit ledger with org-wide reporting

LET'S GO DEEPER

We'd like to walk you through ExhaustShield, live

See the risk router, the redaction layer, and the audit ledger work on real prompts — in a live walkthrough with your team.

> *You essentially pay for intelligence twice — once with money, and again with the proprietary knowledge you reveal.*
>
> **— Satya Nadella, Chairman & CEO, Microsoft**

ExhaustShield exists so your teams keep paying with money only — not with what makes your business yours.

Request a Live Demo